

AI Content Generator

Project Report

Automatically generate Emails, Advertisements, Captions & Product Descriptions using AI

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1. Introduction

In today's digital world, content creation is a daily necessity for businesses, students, and professionals alike. Writing emails, creating social media captions, drafting advertisements, and producing product descriptions all require time, creativity, and consistent quality. Many people, especially those without strong writing skills, find this challenging.

This project presents an AI-powered Content Generator — a web-based tool that takes user input (topic, content type, tone, and word limit) and automatically produces high-quality, relevant written content. The system is built using Flask for the backend and integrates with the Groq API (Llama 3.3 model) for natural language generation.

2. Problem Statement

Manual content creation faces several real-world challenges:

- Time-consuming: Writing good content from scratch takes significant effort.
- Requires creativity and writing skills: Not everyone can write compelling copy.
- Inconsistent quality: Rushed content often lacks polish and professionalism.
- Volume demands: Businesses need large quantities of content regularly.

These challenges lead to delays in communication, poor-quality marketing, and reduced audience engagement. This project directly addresses these pain points by automating content creation with AI.

3. Objectives

- Automatically generate content based on user-provided inputs.
- Support multiple content types: Email, Advertisement, Caption, and Product Description.
- Allow tone selection to match the context (Formal, Casual, Professional, Friendly, Persuasive).
- Maintain quality and relevance in generated output.
- Save time and effort for users with limited writing skills.
- Provide a clean, easy-to-use web interface.

4. Technology Stack

Component	Technology Used
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Backend Framework	Python — Flask
AI / NLP Model	Groq API — Llama 3.3 70B Versatile
Frontend	HTML5, CSS3, Jinja2 Templates
Fonts	Quicksand, Pacifico (Google Fonts)
Environment	python-dotenv (.env file for API key)
IDE	Visual Studio Code
Language	Python 3.14

5. System Design & Architecture

The system follows a simple client-server architecture:

- The user fills in the web form (topic, content type, tone, word limit) and submits it.
- Flask receives the POST request and extracts the form values.
- A structured prompt is built using the inputs and sent to the Groq API.
- The Llama 3.3 model processes the prompt and returns relevant, high-quality content.
- Flask passes the output back to the HTML template and displays it to the user.

The API key is stored securely in a .env file and loaded at runtime using python-dotenv, keeping it out of the source code entirely.

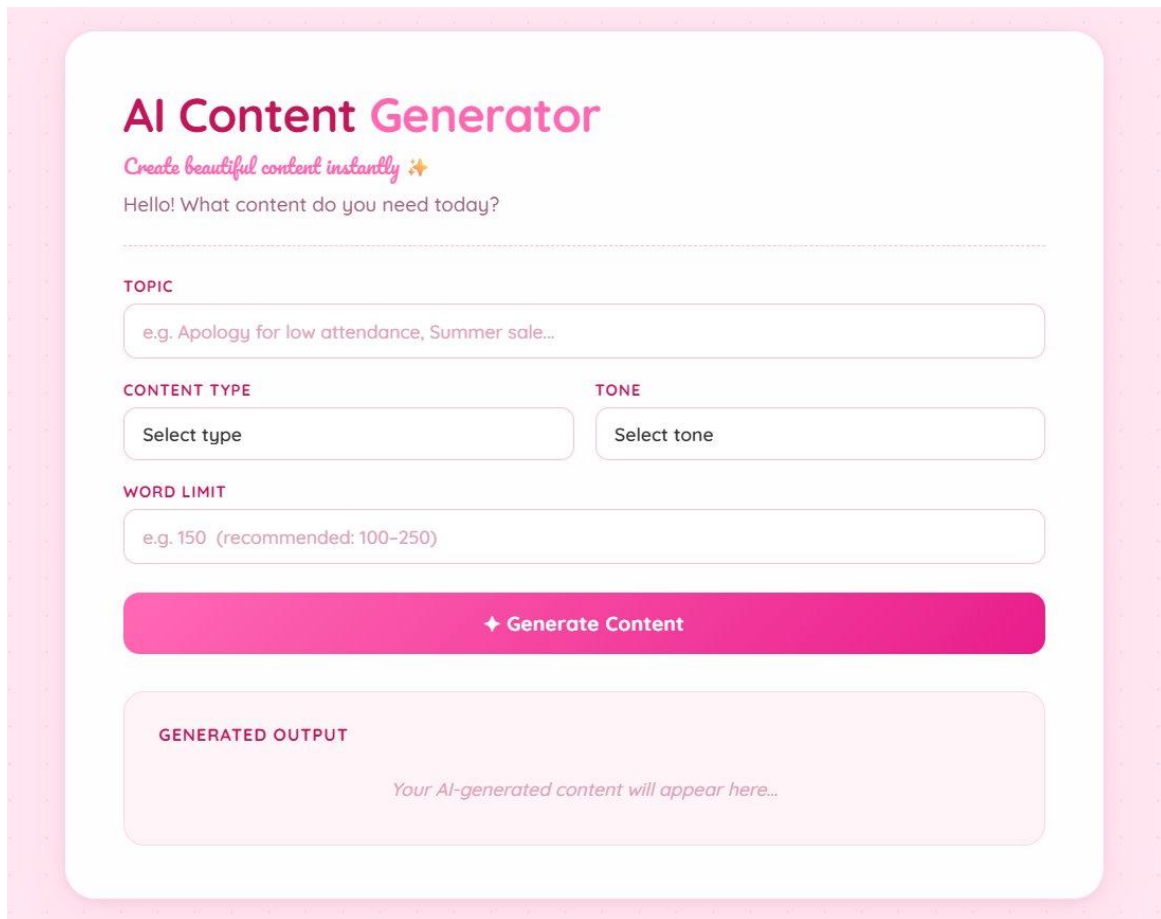
6. Features

- Four content types: Email, Advertisement, Caption, Product Description.
- Five tone options: Formal, Casual, Professional, Friendly, Persuasive.
- Custom word limit control (30–500 words).
- Copy to clipboard button on generated output.
- Error handling with user-friendly messages.
- Secure API key management via .env file.
- Responsive, clean UI with a consistent pink design theme.

7. Screenshots & Output Samples

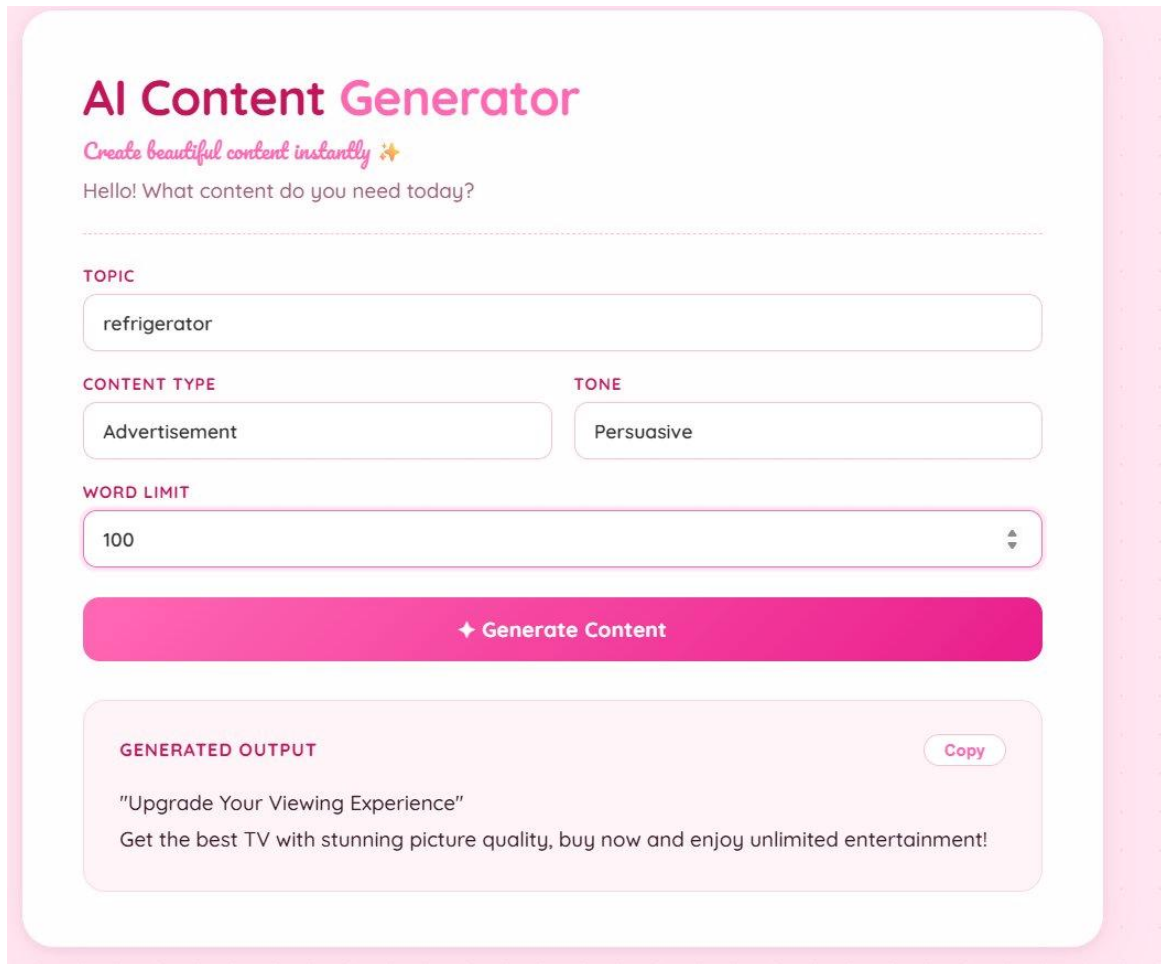
The following screenshots demonstrate the working application across different content types and tones.

7.1 Application Interface (Home Screen)



The screenshot displays the 'AI Content Generator' interface. At the top, the title 'AI Content Generator' is in a bold, dark pink font, followed by the tagline 'Create beautiful content instantly ✨' in a smaller, lighter pink font. Below this is a friendly greeting: 'Hello! What content do you need today?'. The form consists of several input fields: a 'TOPIC' field with a placeholder 'e.g. Apology for low attendance, Summer sale...', a 'CONTENT TYPE' dropdown menu with 'Select type' as the placeholder, a 'TONE' dropdown menu with 'Select tone' as the placeholder, and a 'WORD LIMIT' field with a placeholder 'e.g. 150 (recommended: 100-250)'. A prominent pink button with a white star icon and the text 'Generate Content' is positioned below the input fields. At the bottom, there is a light pink box labeled 'GENERATED OUTPUT' containing the text 'Your AI-generated content will appear here...'.

Figure 1: Main UI — empty state showing the form fields



The image shows a user interface for an AI Content Generator. It has a pink header with the title 'AI Content Generator' and a subtitle 'Create beautiful content instantly ✨'. Below the header is a greeting 'Hello! What content do you need today?'. The form contains three input fields: 'TOPIC' with the value 'refrigerator', 'CONTENT TYPE' with the value 'Advertisement', and 'TONE' with the value 'Persuasive'. There is also a 'WORD LIMIT' dropdown menu set to '100'. A large pink button labeled 'Generate Content' is positioned below the input fields. At the bottom, there is a 'GENERATED OUTPUT' section with a 'Copy' button. The output text reads: 'Upgrade Your Viewing Experience' and 'Get the best TV with stunning picture quality, buy now and enjoy unlimited entertainment!'.

AI Content Generator

Create beautiful content instantly ✨

Hello! What content do you need today?

TOPIC

refrigerator

CONTENT TYPE

Advertisement

TONE

Persuasive

WORD LIMIT

100

Generate Content

GENERATED OUTPUT

Copy

"Upgrade Your Viewing Experience"

Get the best TV with stunning picture quality, buy now and enjoy unlimited entertainment!

Figure 2: UI with selections filled in (Caption, Friendly tone)

7.2 Email Generation

Topic: "Apology for low attendance" | Tone: Formal | Words: 100

AI Content Generator

Create beautiful content instantly ✨

Hello! What content do you need today?

TOPIC

apology for low attendance

CONTENT TYPE

Email

TONE

Formal

WORD LIMIT

100

Generate Content

GENERATED OUTPUT

Copy

Keep Your Food Fresh with Our Refrigerator

Stay healthy with our advanced refrigerator, featuring automatic temperature control and spacious storage. Enjoy fresh food, reduced energy bills, and a sleek design. Upgrade your kitchen today and experience the difference. Call now to get a special discount!

Figure 3: Formal email generated for apology for low attendance

7.3 Advertisement Generation

Topic: "Refrigerator" | Tone: Persuasive | Words: 100

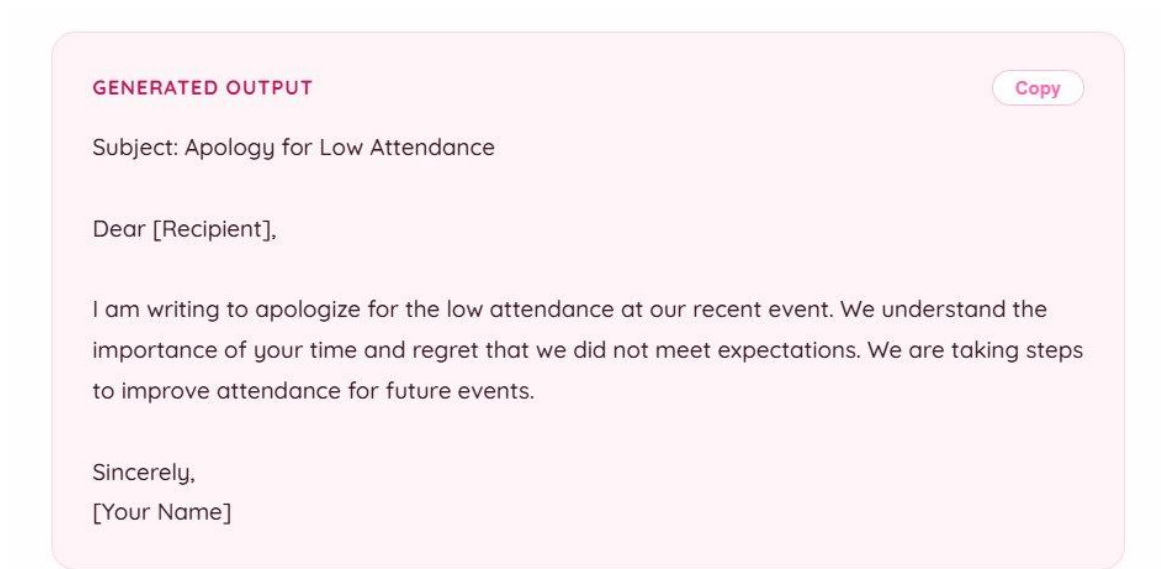


Figure 4: Persuasive advertisement for a refrigerator

7.4 Product Description Generation

Topic: "Mobile phone" | Tone: Professional | Words: 100

AI Content Generator

Create beautiful content instantly ✨

Hello! What content do you need today?

TOPIC

mobile phone

CONTENT TYPE

Product Description

TONE

Professional

WORD LIMIT

100

✦ Generate Content

GENERATED OUTPUT

Copy

Experience unparalleled connectivity with our mobile phone, featuring a high-resolution display, advanced camera system, and long-lasting battery life. Enjoy seamless navigation, lightning-fast processing, and ample storage for your apps and media. Stay connected, capture life's moments, and access a world of entertainment on the go.

Figure 5: Professional product description for a mobile phone

7.5 Caption Generation

Topic: "Couple photo both hug each other" | Tone: Friendly | Words: 50

AI Content Generator

Create beautiful content instantly ✨

Hello! What content do you need today?

TOPIC

couple photo both hug each other

CONTENT TYPE

Caption

TO NE

Friendly

WORD LIMIT

50

✦ Generate Content

Figure 6: Caption generated with hashtags for a couple photo

7.6 Output Detail Views

GENERATED OUTPUT

Copy

Keep Your Food Fresh with Our Refrigerator

Stay healthy with our advanced refrigerator, featuring automatic temperature control and spacious storage. Enjoy fresh food, reduced energy bills, and a sleek design. Upgrade your kitchen today and experience the difference. Call now to get a special discount!

Figure 7: Caption output with hashtags — close-up view

GENERATED OUTPUT

Copy

Love is in the air. Couple embracing each other, filled with joy and happiness #CoupleGoals #LoveOfMyLife #HugMeTight

Figure 8: Refrigerator advertisement output — close-up view

8. How It Works

Step-by-step flow of the application:

1. User opens the web interface at `http://127.0.0.1:5000`.
2. User enters a topic, selects content type, tone, and word limit.
3. On clicking "Generate Content", the form data is sent to Flask via POST request.
4. Flask builds a detailed prompt incorporating all inputs and sends it to the Groq API.
5. The Llama 3.3 70B model returns relevant, human-quality content.
6. The output is displayed on the same page with a Copy button for convenience.

9. Challenges Faced

- Initial use of `distilgpt2` model produced random, incoherent outputs. Solved by switching to the Groq API with the Llama 3.3 instruction-following model.
- The `llama3-8b-8192` model was decommissioned mid-development. Updated to `llama-3.3-70b-versatile` which resolved the issue.
- API key management: Learned to use `.env` files with `python-dotenv` to keep credentials secure and out of source code.
- Flask app variable not defined error: Corrected the order of declarations (`app = Flask(__name__)` must precede all route definitions).

10. Conclusion

This project successfully demonstrates how AI can be integrated into a practical web application to solve a real-world problem. The AI Content Generator is capable of producing relevant, high-quality content across multiple categories with different tonal styles — tasks that would otherwise take significant time and skill.

The project helped develop hands-on skills in Python web development (Flask), API integration, prompt engineering, frontend design, and secure coding practices. It reflects a complete development cycle: from identifying a problem to designing, building, debugging, and delivering a working solution.

— End of Report —
